

Beriberi

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What is beriberi?

Thiamine is an essential nutrient that people obtain through the diet. It plays a role in many bodily functions, including:

- ☐ breaking down carbohydrate
- ☐ muscle contraction
- ☐ conduction in nerve cells
- ☐ creating glucose
- ☐ creating acids that help digestion

The word beriberi comes from a Sinhalese word meaning "extreme weakness," as it can cause severe and even life-threatening symptoms. Beriberi is a serious and potentially life-threatening condition that develops if a person has a severe thiamine deficiency.

severe thiamine deficiency may prevent the body from performing these functions properly

o There are two main forms of beriberi:

1-Wet beriberi

which mainly affects the cardiovascular system, causing poor circulation and fluid buildup in the tissues.

2-Dry beriberi

which primarily affects the nervous system, leading to the degeneration of the nerves. Degeneration typically begins in the legs and arms and may lead to muscle atrophy and loss of reflexes

Causes:

The leading cause of beriberi is either a diet low in thiamine or a problem that limits the body's ability to process thiamine.

people who abuse alcohol will develop thiamine deficiency. Alcohol makes it more difficult for the body to process and absorb thiamine.

Babies may also develop beriberi if the breast milk that they drink is lacking in thiamine or if they only drink formula without this vitamin.

Other people who may have a higher risk of beriberi include:

- ❖ older adults
- ❖ people with diabetes
- ❖ people with HIV
- ❖ people who have had
- ❖ bariatric surgery

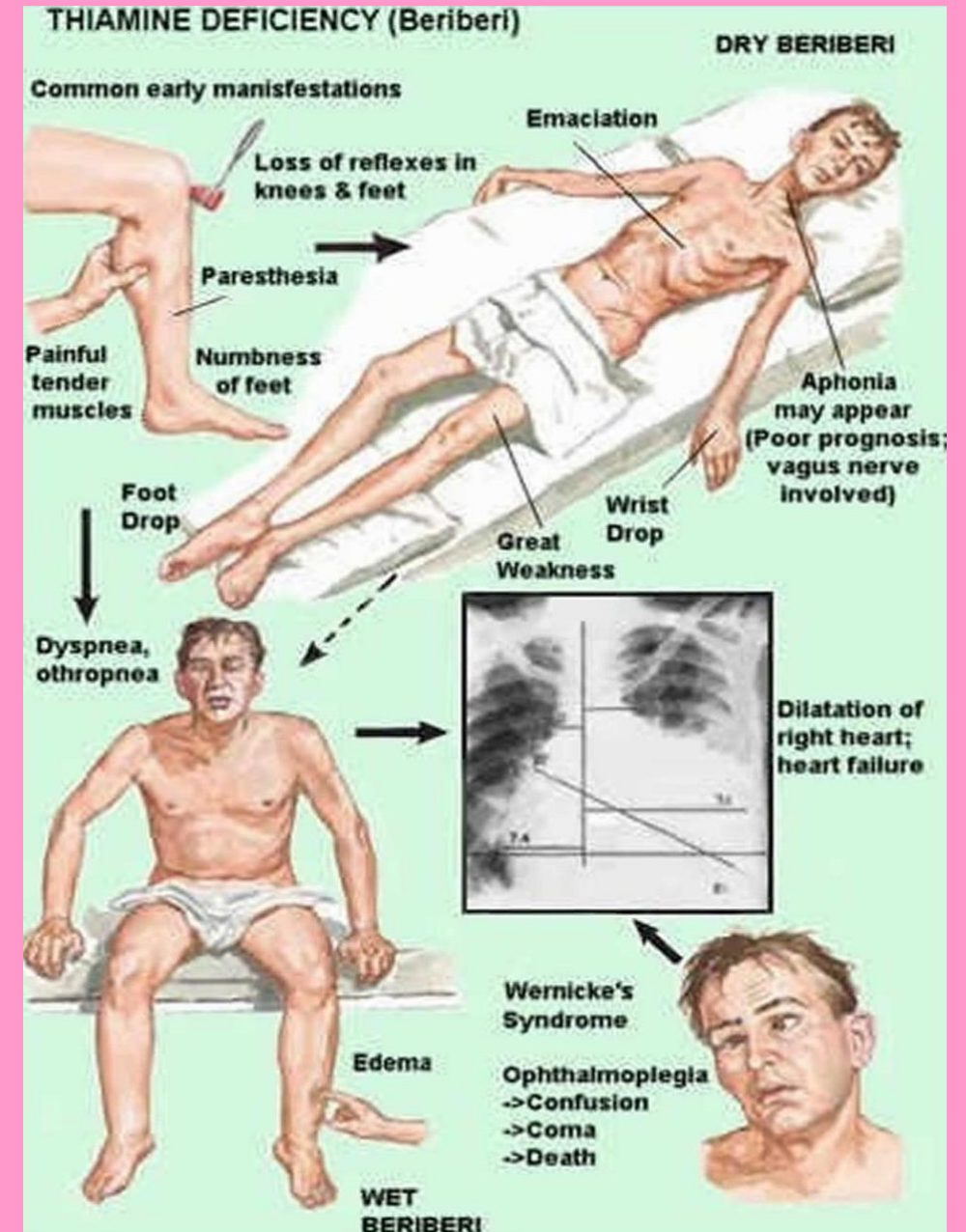


Clinical Manifestations:

- Neurologic Symptoms
- Poor memory, irritability, sleep disturbance
- Paresthesia
- Muscle cramps
- Decreased vibratory position sensation
- Absent knee and ankle jerk
- Muscle atrophy
- Foot drop
- Cardiovascular Symptoms
- Tachycardia
- Chest pain
- Wide pulse pressure:
- Heart failure
- Hypotension shock

Other Manifestations:

- Anorexia
- Abdominal Discomfort
- Constipation
- dysphagia



Symptoms:

The symptoms of beriberi may vary depending on its type.

Symptoms of wet beriberi include:

- increased heart rate
- severe lack of energy or constant fatigue
- shortness of breath
- waking at night due to shortness of breath

Symptoms of dry beriberi include:

- general pain and body aches
- vomiting
- difficulty walking
- confusion
- numbness in the hands or feet
- paralysis in the lower legs

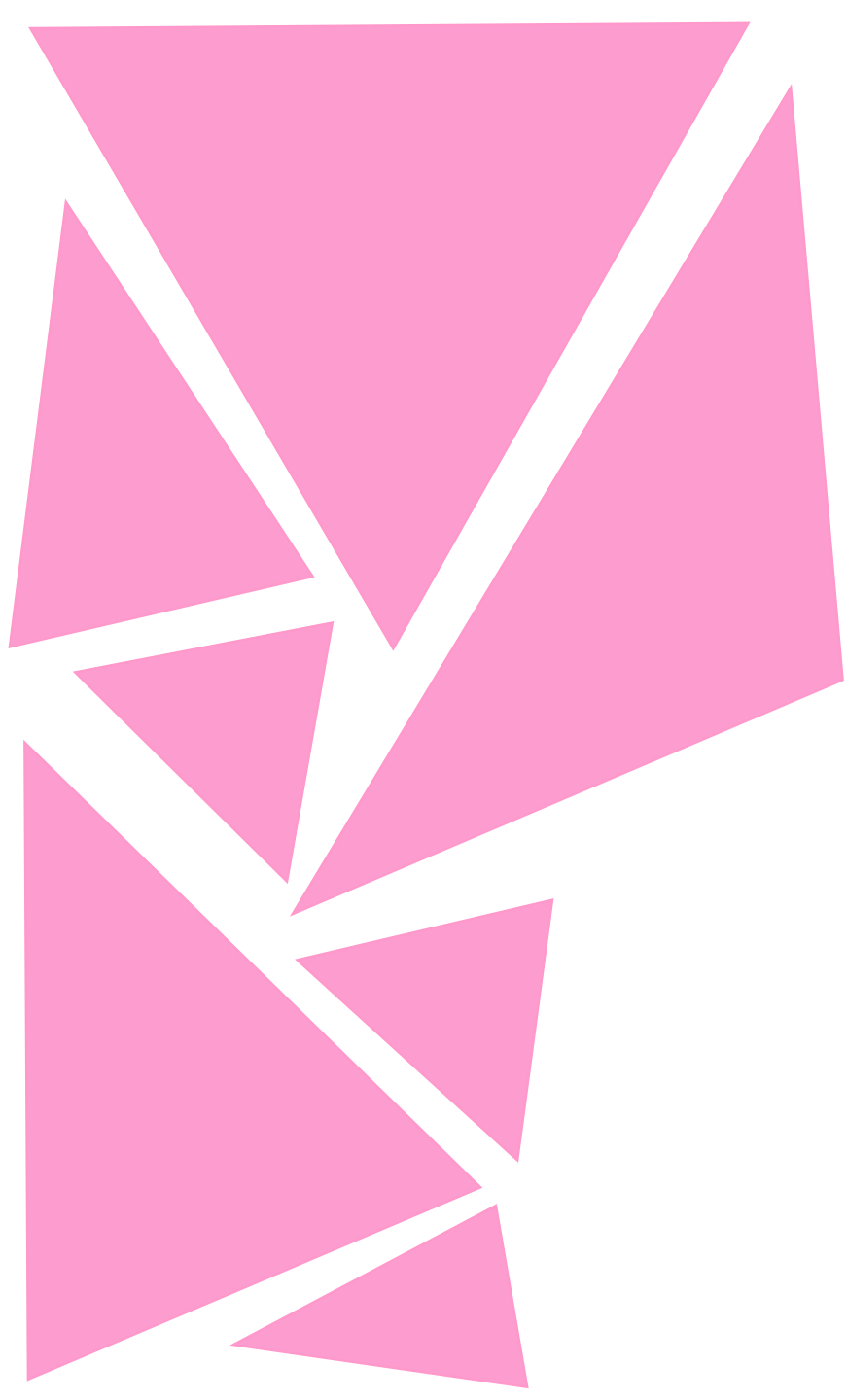
Pathophysiology

Dry Beriberi

- Poor caloric intake and Relative Physical Inactivity
- Degeneration of the myelin the muscular sheaths

Wet Beriberi

- Peripheral Vasodilation occurs → High cardiac output → salt and water retention (RAAS)
 - ↙ Edema
 - ↘ Heart increases workload



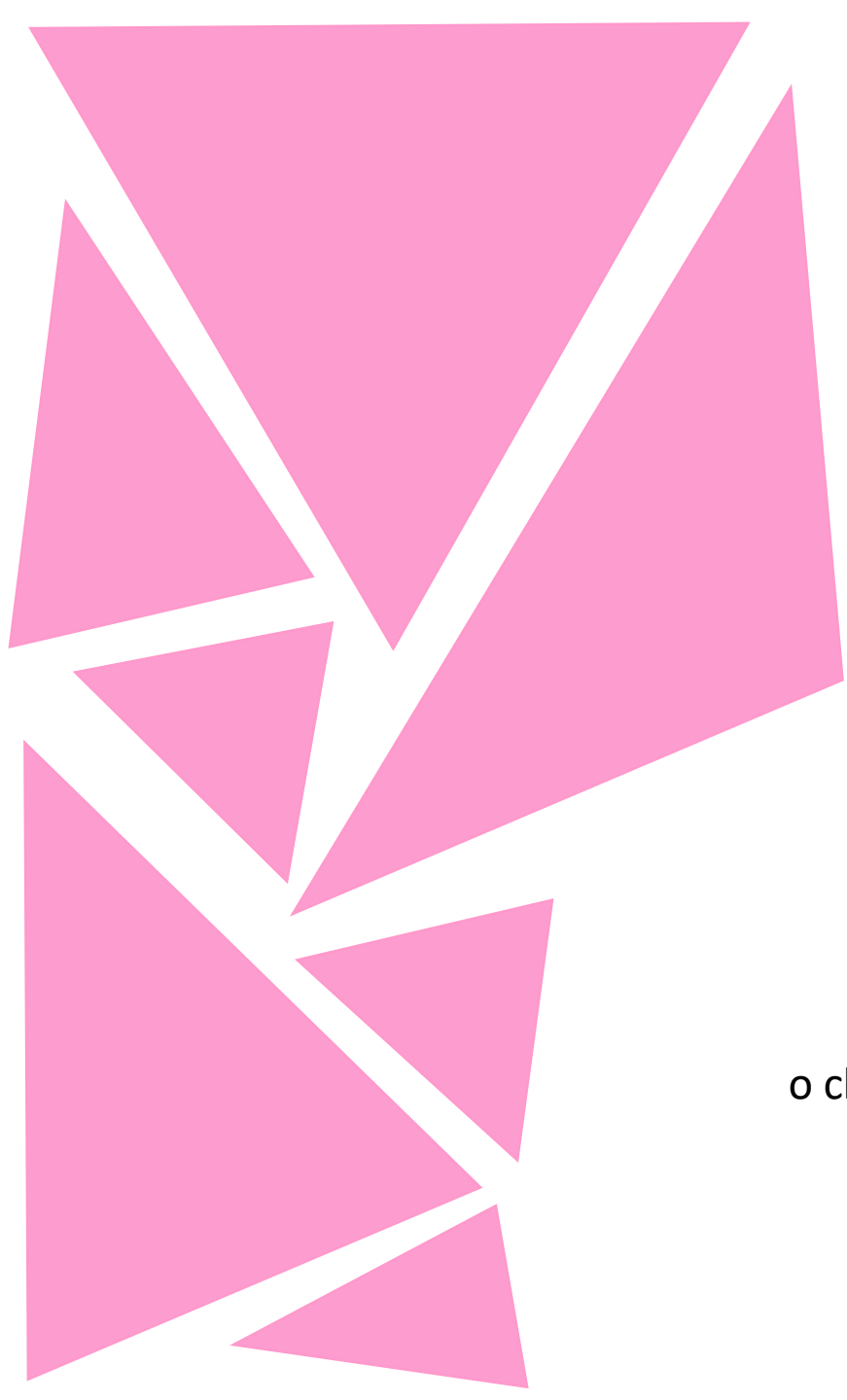
Diagnosis

Diagnostic and Lab Findings.

Blood and Urine Tests will measure levels of Vitamin B1 specifically, erythrocyte transketolase activity.

Positive to beriberi: low concentration of thiamine in blood and high in urine.

o blood and urine tests to help measure the levels of thiamine in a person bloodstream to diagnose beriberi

A decorative graphic on the left side of the slide, consisting of several overlapping pink triangles of various sizes, creating a geometric, abstract pattern.

Nursing Diagnosis

Imbalance Nutrition: Less than body requirements Activity Intolerance

- perform a physical exam to check for signs of other neurological damage or issues with the heart.
- Signs of neurological damage can include difficulty walking or staying balanced, lack of coordination, and weak reflexes.
- check a person's heart rate and breathing rate and look for swelling in the lower limbs, which can indicate heart problems.

Treatment

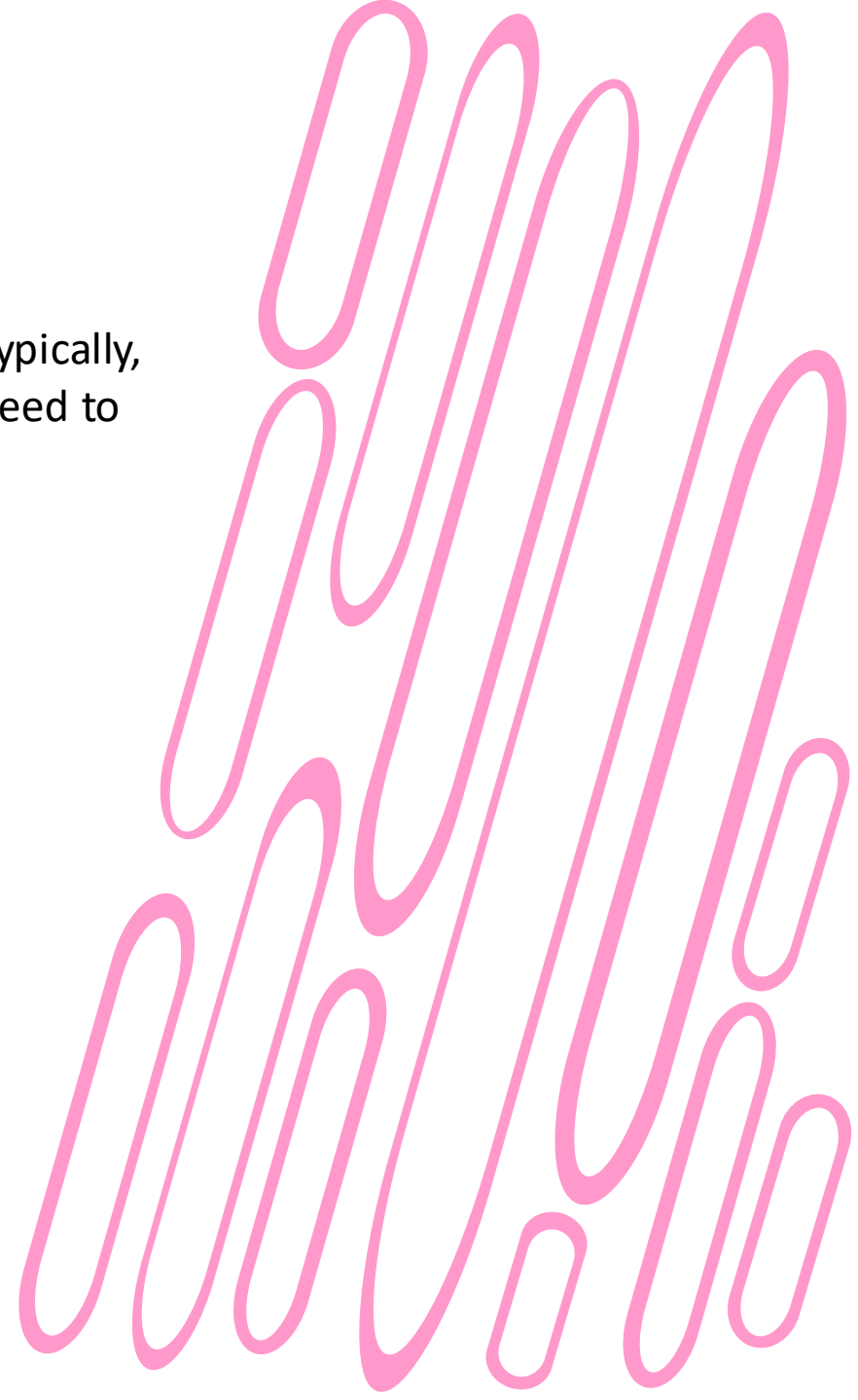
The goal of treatment for beriberi is to increase thiamine levels in the body.

Prompt administration of parenteral thiamine is indicated. (The recommended dose is 50 mg given IM for several days and maintenance is 2.5 - 5mg per day if malabsorption is suspected)

- Doctors may recommend oral supplements or injections to deliver this thiamine, depending on a person's overall health. They may also suggest taking other supplements to support treatment.
- treat any complications
- check a person's heart rate and breathing rate and look for swelling in the lower limbs, which can indicate heart problems.
- regular blood tests to check the person's thiamine levels until they return to normal

Prevention

- Preventing beriberi requires a person to consume enough thiamine in their diet. Typically, a person who eats a healthful, balanced diet with a focus on whole foods will not need to worry about thiamine deficiency.
- The Department of Health and Human Services as Trusted Source note that eating fortified bread and cereals is the most common way for people in the U.S. to reach the recommended daily intake of thiamine
- Pregnant or breastfeeding women should pay close attention to their thiamine consumption. Additionally, anyone who is using formula to feed an infant should check to see that it contains enough thiamine.



Top 10 Foods Highest in Thiamin (Vitamin B1)

1mg of Thiamin = 100% of the Daily Value (%DV)

1 Lean Pork Chops



96% DV (1.1mg)
in a 6oz chop

332 calories

2 Fish (Salmon)



48% DV (0.6mg)
per 6oz fillet

350 calories

3 Flax Seeds



39% DV (0.5mg)
per oz

152 calories

4 Navy Beans



36% DV (0.4mg)
per cup

255 calories

5 Green Peas



35% DV (0.4mg)
per cup cooked

134 calories

6 Firm Tofu



33% DV (0.4mg)
per cup

363 calories

7 Brown Rice



30% DV (0.4mg)
per cup

248 calories

8 Acorn Squash



29% DV (0.3mg)
per cup cooked

115 calories

9 Asparagus



24% DV (0.3mg)
per cup cooked

40 calories

10 Mussels



21% DV (0.3mg)
per 3oz

146 calories

HIGH THIAMINE (B1) FOODS



pork loin (lean)
1.4 mg/120 g serve



leg ham
0.62 mg/150 g serve



salmon
0.61 mg/64 g serve



pili nuts
0.52 mg/48 g serve



oatmeal (steel cut)
0.35 mg/73 g serve



macadamia nuts
0.35 mg/29 g serve



liver
0.3 mg/77 g serve



whole wheat bread
0.28 mg/68 g serve



sunflower seeds
0.26 mg/17 g serve



brown rice
0.26 mg/115 g serve



lentils
0.25 mg/34 g serve



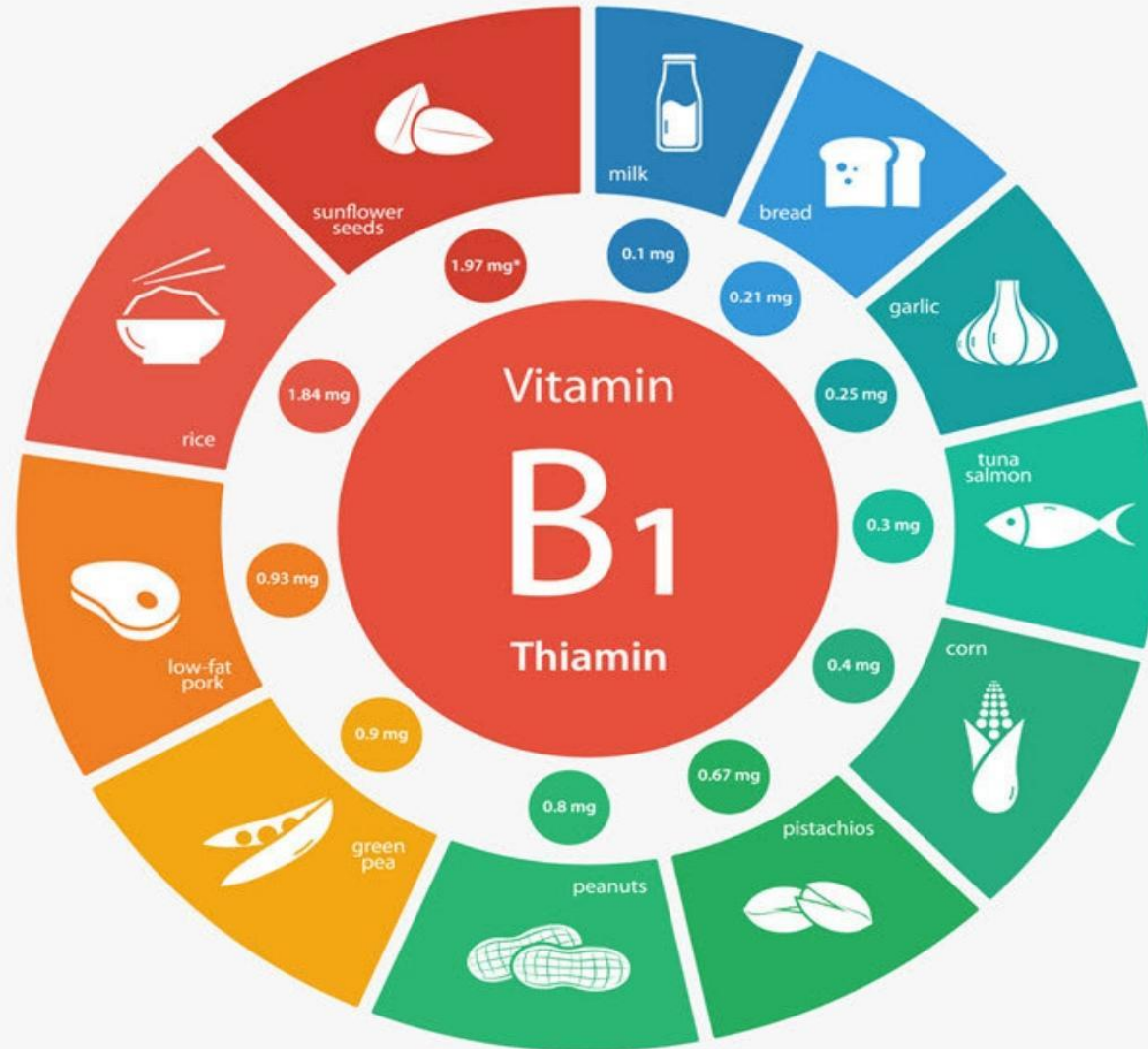
peanuts
0.25 mg/37 g serve



OPTIMISING
NUTRITION

serving size = average consumed by Optimisers
optimal nutrient intake = 3.0 mg/2000 cal
bliss point (minimum) = 1.3 mg/2000 cal

Foods high in vitamin B₁



* vitamin content, mg/100gr

Avviva,F ,. et al.(2011). Thiamin (Vitamin B1)

World health organization.(1999).Thiamine Deficiency.

Mayank , M.(2020).Thiamine deficiency disorders: a clinical perspective.

Tognazzi, S ,.et al .(2022).Thiamine deficiency: A never ending story from the pediatric age till adulthood, two cases and review of diagnosis and treatment

Thank You